

[0] ĐỌC CODE TRƯỚC (bắt buộc)

bash

```
find . -type f \( -name "*.js" -o -name "*.ts" \) | grep -v node_modules | sort
cat package.json
cat src/index.js 2>/dev/null || cat src/app.js 2>/dev/null || cat src/server.js
ls src/routes/ && cat src/routes/*.js 2>/dev/null | head -200
cat src/controllers/*.controller.js 2>/dev/null | head -200
cat src/config/db.js 2>/dev/null || cat src/database/connection.js
cat src/utis/response.js 2>/dev/null || cat src/helpers/response.js
cat src/middlewares/auth.middleware.js 2>/dev/null || cat src/middlewares/auth.js
```

→ Ghi nhớ: framework, ORM/raw-query, response format, auth middleware, file naming convention.

[1] CONVENTIONS (áp dụng toàn bộ)

- **Response:** {success:bool, data?, message?, pagination?}
- **Pagination:** params page(def:1) + limit(def:20) → {current_page, per_page, total_items, total_pages}
- **Auth:** Bearer JWT → req.user = {user_id, roles:string[]}
- **Seller store:** SELECT store_id FROM user_stores WHERE user_id=@uid AND store_member_role='OWNER' AND is_active=1
- **Soft delete:** set deleted_at=SYSUTCDATETIME() hoặc status='INACTIVE'
- **SQL Server:** dùng @param, OFFSET x ROWS FETCH NEXT y ROWS ONLY
- **Comment code:** tối thiểu — chỉ comment logic không tự giải thích được

GROUP 10: Orders – Customer (34–37)

34 · Place Order **POST** /api/v1/orders ⚠ TRANSACTION

Body: {store_id, items:[{product_id,quantity}], recipient_name, recipient_phone, shipping_address_line, shipping_province, shipping_ward?, shipping_district?, payment_method(COD|BANK_TRANSFER), customer_note?, address_id?, voucher_code?}

Transaction steps:

1. Verify tất cả product_id thuộc store_id, status='ACTIVE'
2. Check stock_quantity >= quantity từng product
3. subtotal = SUM(price * qty)

4. Nếu `voucher_code`: validate → tính `discount_amount` (check `min_order_amount`, `usage_limit`, `ends_at` ≥ NOW(), per-customer limit)
5. `total_amount = subtotal - discount + shipping_fee` (=0 mặc định)
6. INSERT orders (`order_code`='ORD-{yyyyMMdd}-{nanoid(8)}', `order_status`='PENDING')
7. INSERT order_items (`product_name_snapshot`, `product_image_url_snapshot`, `unit_price`=p.price)
8. UPDATE products SET `stock_quantity`=qty, `sold_quantity`+=qty
9. INSERT order_status_histories (prev=NULL, new='PENDING', changed_by=@uid)
10. Nếu voucher: INSERT order_vouchers; UPDATE vouchers SET `used_count`+=1
11. Xóa cart_items tương ứng (nếu đặt từ giỏ hàng)

→ 201: {`order_id`, `order_code`, `order_status`:'PENDING', `total_amount`} → 400 không đủ hàng | 400 voucher lỗi

35 · List My Orders GET /api/v1/orders

Params: `status?`, `page`, `limit`

sql

```
SELECT o.order_id, o.order_code, o.order_status, o.total_amount, o.created_at,
       s.store_name, oi.product_name_snapshot AS first_item
FROM orders o
JOIN stores s ON o.store_id=s.store_id
OUTER APPLY (SELECT TOP 1 product_name_snapshot FROM order_items WHERE
order_id=o.order_id) oi
WHERE o.customer_id=@uid [AND o.order_status=@s]
ORDER BY o.created_at DESC
```

→ 200: {`orders`[], `pagination`}

36 · Get Order Detail GET /api/v1/orders/:id

- Verify `order.customer_id` = @uid
- JOIN: orders + order_items + stores + shipments (LEFT)

→ 200: {...order, `items`[], `store`{name,logo}, `shipment`?{`status`,`tracking_code`}} → 403 | 404

37 · Cancel Order POST /api/v1/orders/:id/cancel

Body: {`cancel_reason`?}

- Verify ownership
- Check `order_status` = 'PENDING' → nếu không: 400 "Không thể hủy đơn đã xử lý"

- `UPDATE orders SET order_status='CANCELLED', cancelled_by_user_id=@uid, cancel_reason=@r, cancelled_at=NOW()`
- `INSERT order_status_histories (prev='PENDING', new='CANCELLED')`
- Hoàn stock: `UPDATE products SET stock_quantity+=qty` theo từng order_item

→ **200**: {message: "Đã hủy đơn"} | **400** | **403** | **404**

CHECKLIST:

- [] Tất cả 4 route (34-37) đăng ký đúng index router.
- [] Auth middleware đúng role 'Customer'.
- [] **BẮT BUỘC** có Transaction cho API #34 (Place Order) để đảm bảo tính toàn vẹn (tạo đơn, trừ kho, lưu lịch sử).
- [] Logic hoàn kho chính xác khi hủy đơn hàng (API 37).
- [] Parameterized query được sử dụng.
- [] Response format nhất quán.